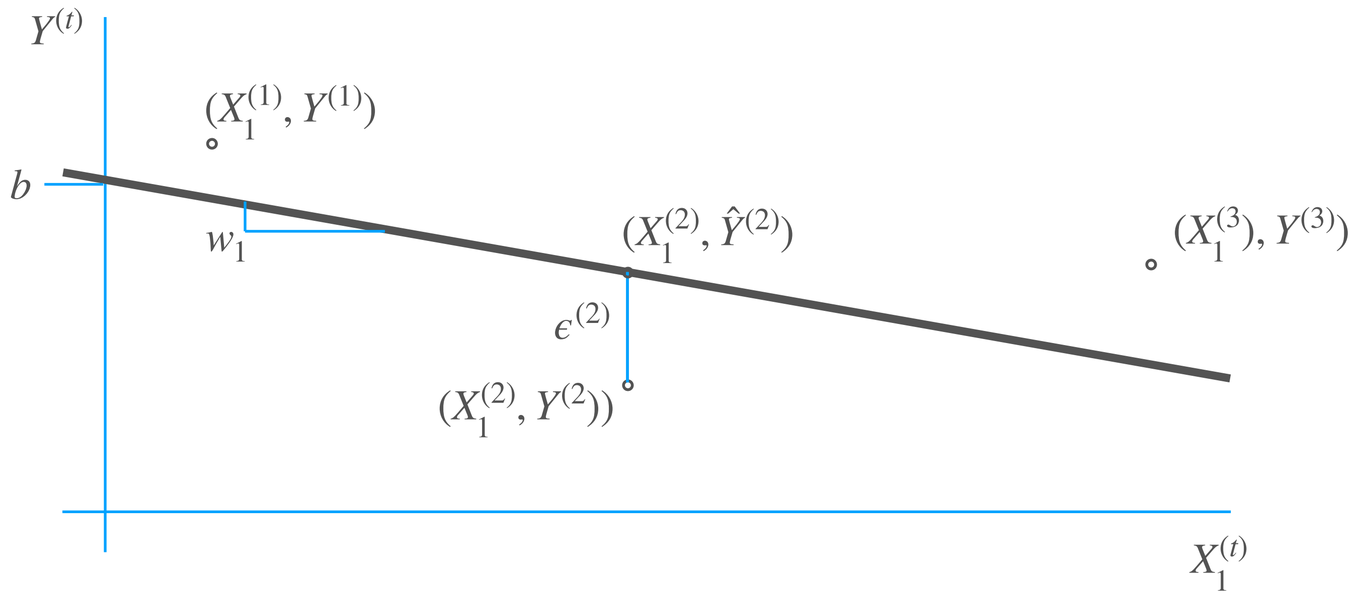
Reminder

- HW1 due tomorrow

Linear regression



- Regression = supervised prediction of *real* labels
  - data: features $X^{(t)} \in \mathbb{R}^d$, labels $Y^{(t)} \in \mathbb{R}$, $t = 1 : T$
- *Linear* regression = predict a linear function of features
  - model: predict $\hat{Y}^{(t)} = w \cdot X^{(t)} = \sum_i w_i X_i^{(t)}$ or $w \cdot X^{(t)} + b$
  - can omit b if we have a constant feature

Linear regression

- Objective:
  - minimize sum of squared errors:

$$\frac{1}{2}\sum_{t=1}^{T}(\underbrace{Y^{(t)} - \hat{Y}^{(t)}}_{\epsilon^{(t)}})^2$$

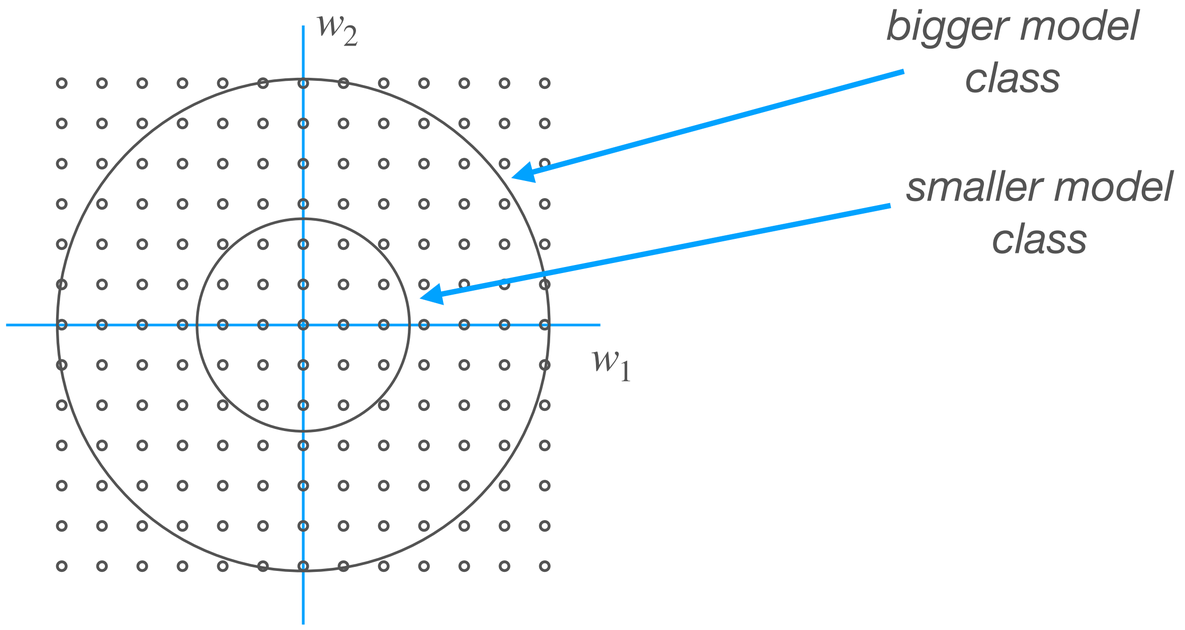
  - plus optional regularizer (see next slides)
- E.g., minimize

$$L(w) = \frac{1}{2}\sum_{t=1}^{T}(Y^{(t)} - \hat{Y}^{(t)})^2 + \frac{\lambda}{2}\|w\|_2^2$$

Model complexity

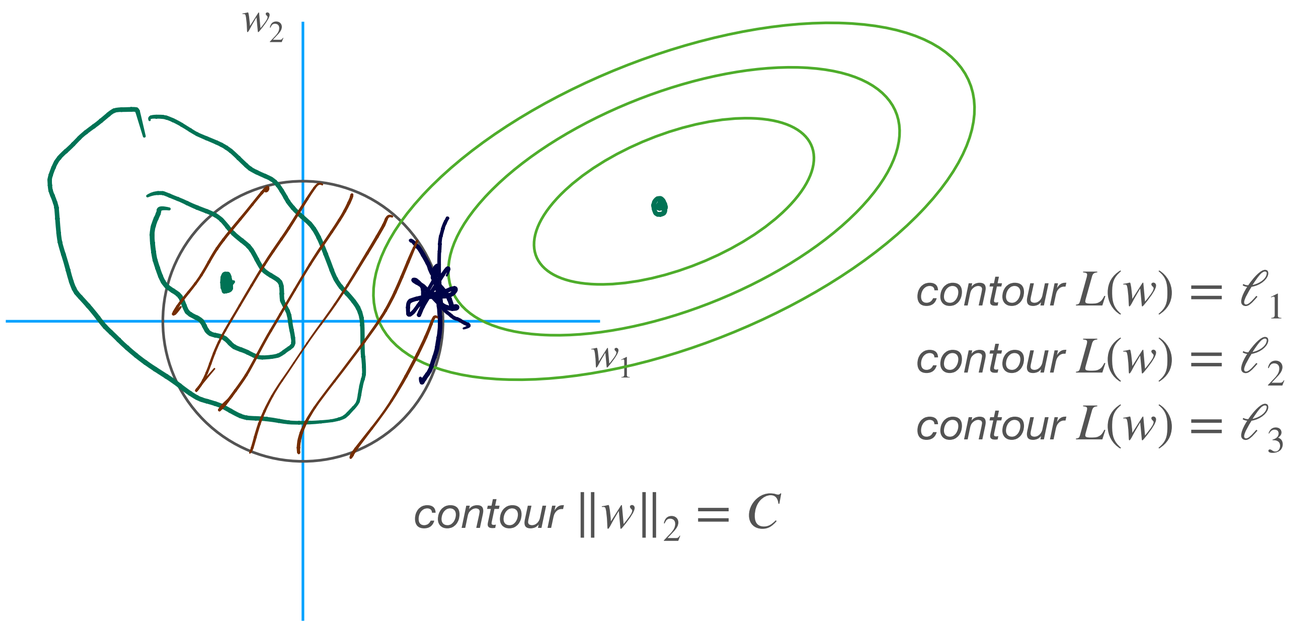
*Terminology: **model** can mean either the structure or the structure plus fitted parameters*
*To distinguish, **model class** vs. **trained/fitted model***

Trained model = hypothesis
One or more model classes make up hypothesis space



- Model complexity (for model selection):
  - how many features/weights
  - how many *nonzero* weights
  - *size* of weights (there are more effectively-distinct weight vectors in $\{w \mid \|w\| \leq 100\}$ than in $\{w \mid \|w\| \leq 1\}$)

Model selection

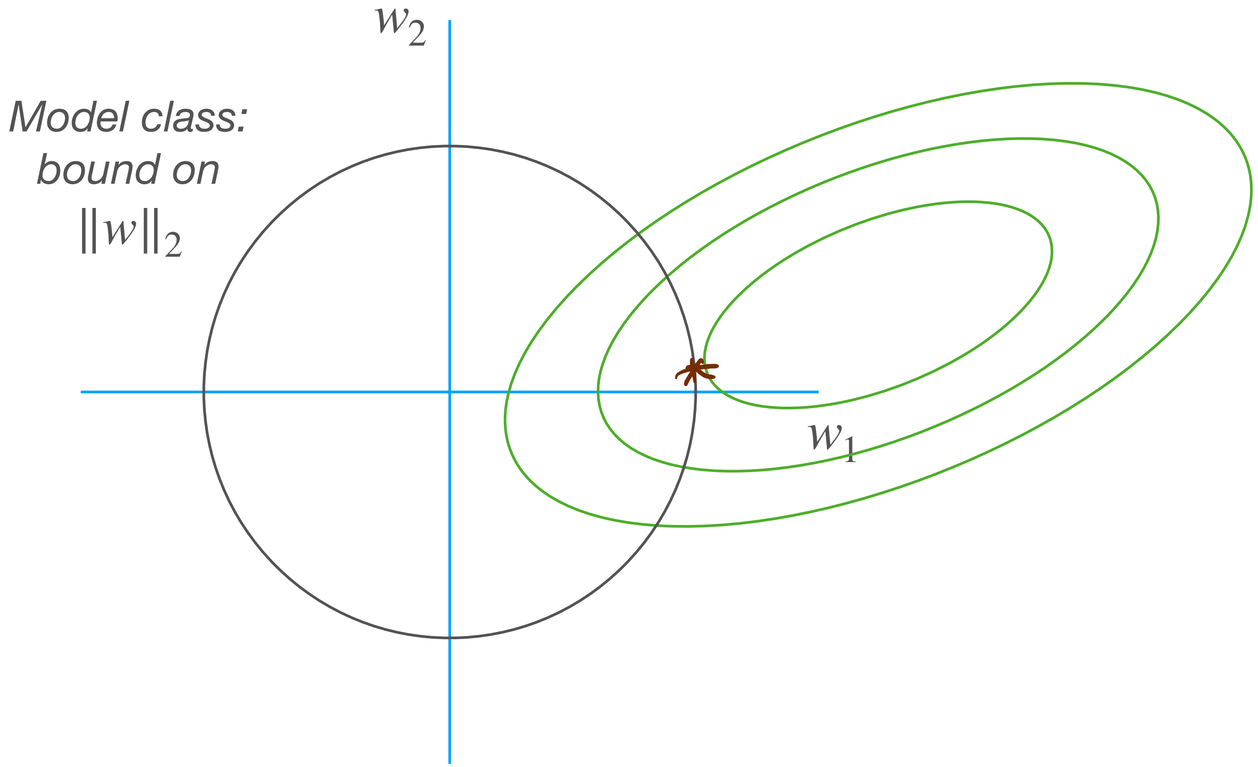


- To limit complexity, can limit (e.g.) norm of weight vector
- Corresponding model fitting problem:

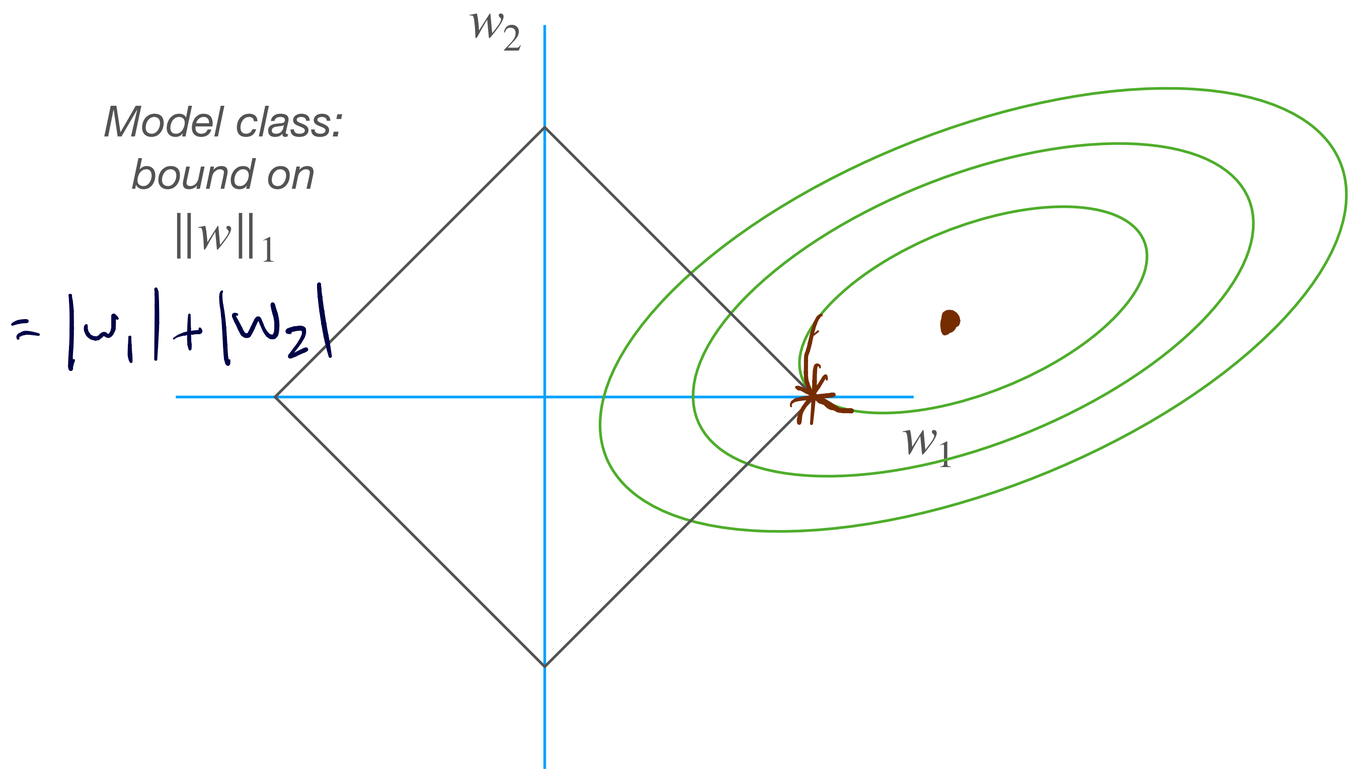
$$\min_w L(w) = \frac{1}{2}\sum_{t=1}^{T}(Y^{(t)} - X^{(t)} \cdot w)^2 \quad \text{s.t.} \quad \|w\|_2 \leq C$$

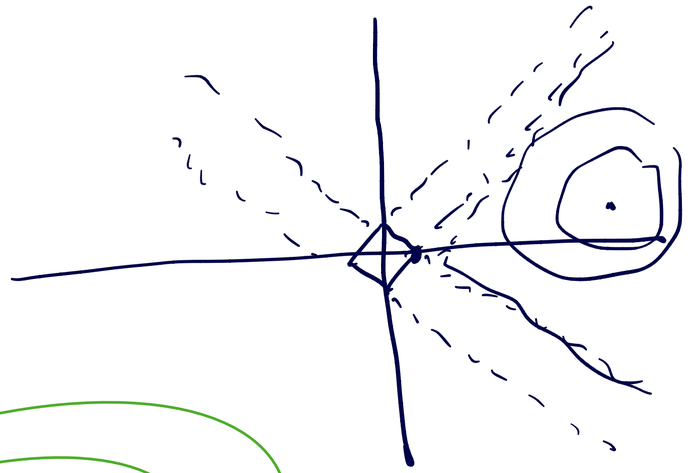
- Model selection = picking C

Size (norm) and count of nonzeros are related

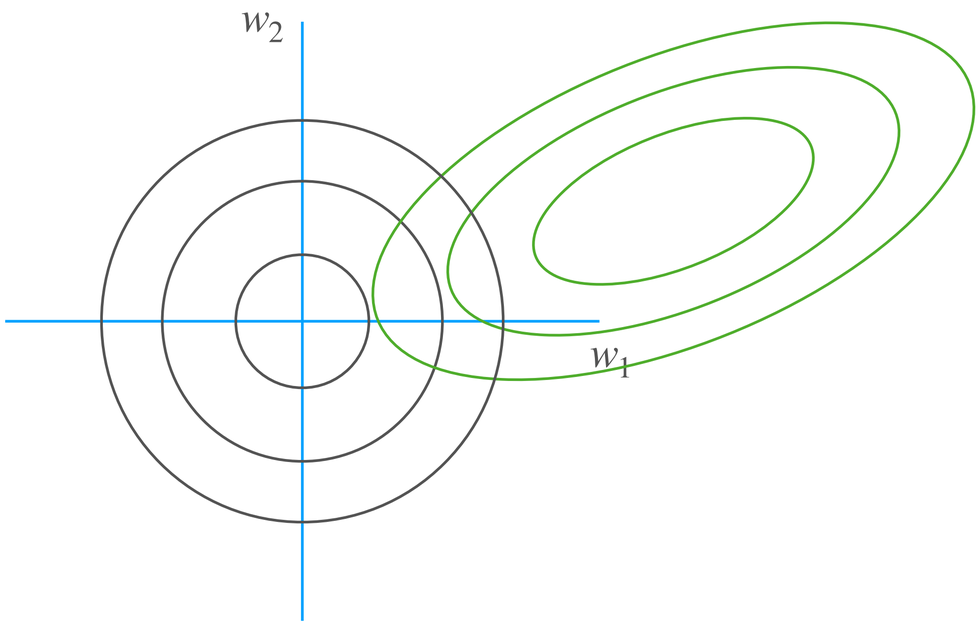


Size (norm) and count of nonzeros are related





Regularization by penalty vs. constraint



- Instead of constrained minimization

$$\min_w L(w) = \frac{1}{2}\sum_{t=1}^{T}(Y^{(t)} - X^{(t)} \cdot w)^2 \quad \texttt{s.t.} \quad \|w\|_2 \leq C$$

- we can try to directly trade off $L(w)$ and $\|w\|_2$ in optimizer (avoid outer model selection loop):

$$\min_w L(w) = \frac{1}{2}\sum_{t=1}^{T}(Y^{(t)} - X^{(t)} \cdot w)^2 + \frac{\lambda}{2}\|w\|_2^2$$

Bias-variance tradeoff

- Regularization parameters λ, C trade off underfitting vs. overfitting
  - underfitting is often called ***bias****:* the correct answer is not an option, so we pick something that might be far from it
  - overfitting shows up as ***variance*** of parameters w: more choices $\rightarrow$ more variability in which one we select

regression : SSE = bias^2 + variance

Overfitting, bias, variance example

```
n = 10
k = 7
X = np.random.normal(size=(n, k))
Y = np.random.normal(size=(n, 1))
w, _, _, _ = np.linalg.lstsq(X, Y, rcond=None)
(np.linalg.norm(w), np.mean((Y - X@w)**2))
```

- What is the smallest MSE this code should return if there were no selection bias? $\rightarrow 1$
- What is the actual best w? $\rightarrow w = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$

Overfitting, bias, variance example

```
n = 10
k = 7
X = np.random.normal(size=(n, k))
Y = np.random.normal(size=(n, 1))
w, _, _, _ = np.linalg.lstsq(X, Y, rcond=None)
(np.linalg.norm(w), np.mean((Y - X@w)**2))
```

- What is the smallest MSE this code should return if there were no selection bias?
- What is the actual best w?

```
# three runs (n=10, k=7):

(1.729258520013365, 0.09403950328236421)
(1.9706932994777069, 0.07379394277037196)
(3.591128626783351, 0.47238275606461944)
```

Overfitting, bias, variance example

```
n = 10
k = 7
X = np.random.normal(size=(n, k))
Y = np.random.normal(size=(n, 1))
w, _, _, _ = np.linalg.lstsq(X, Y, rcond=None)
(np.linalg.norm(w), np.mean((Y - X@w)**2))
```

- What is the smallest MSE this code should return if there were no selection bias?
- What is the actual best w?

```
# three runs (n=10, k=7):

(1.729258520013365, 0.09403950328236421)
(1.9706932994777069, 0.07379394277037196)
(3.591128626783351, 0.47238275606461944)

# three runs (n=20, k=2):

(0.20006773436081007, 0.8932440732695813)
(0.3765035073938464, 0.9200354120567253)
(0.5764201246357761, 0.7614382010070438)
```

Linear regression: summary of objective

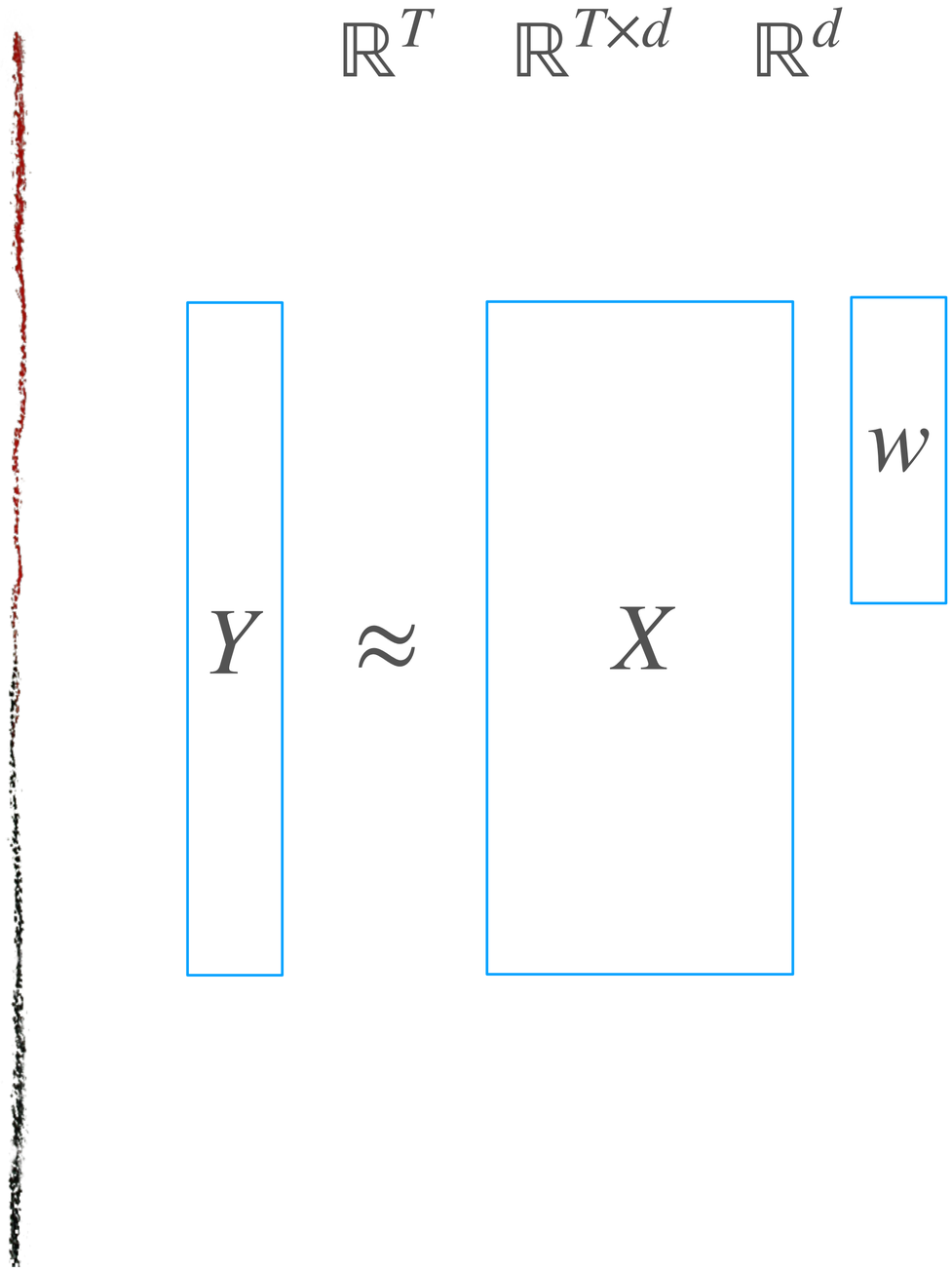
- Objective:
  - minimize sum of squared errors: $\frac{1}{2}\sum_{t=1}^{T}(Y^{(t)} - \hat{Y}^{(t)})^2$
  - plus optional regularizer: e.g., L_2, L_1, both
  - $\|w\|_2^2$ called *ridge regression*
  - $\|w\|_1$ called *LASSO*
  - if we use both, *elastic net*
- E.g., minimize

$$L(w) = \frac{1}{2}\sum_{t=1}^{T}(Y^{(t)} - \hat{Y}^{(t)})^2 + \frac{\lambda}{2}\|w\|_2^2 + \beta\|w\|_1$$

Solving it

- To find best w, several options:
  - minimize symbolically by setting $\nabla_w L$ to 0
    - sub-choices: QR, SVD, Cholesky
  - iterative optimizer
    - sub-choices: stochastic vs. full gradient
    - 1st-order like SGD or Adam
    - 2nd-order like Newton
    - in-between like L-BFGS
    - tools like momentum, rescaling
  - choice of covariance form or kernel form

Schematic

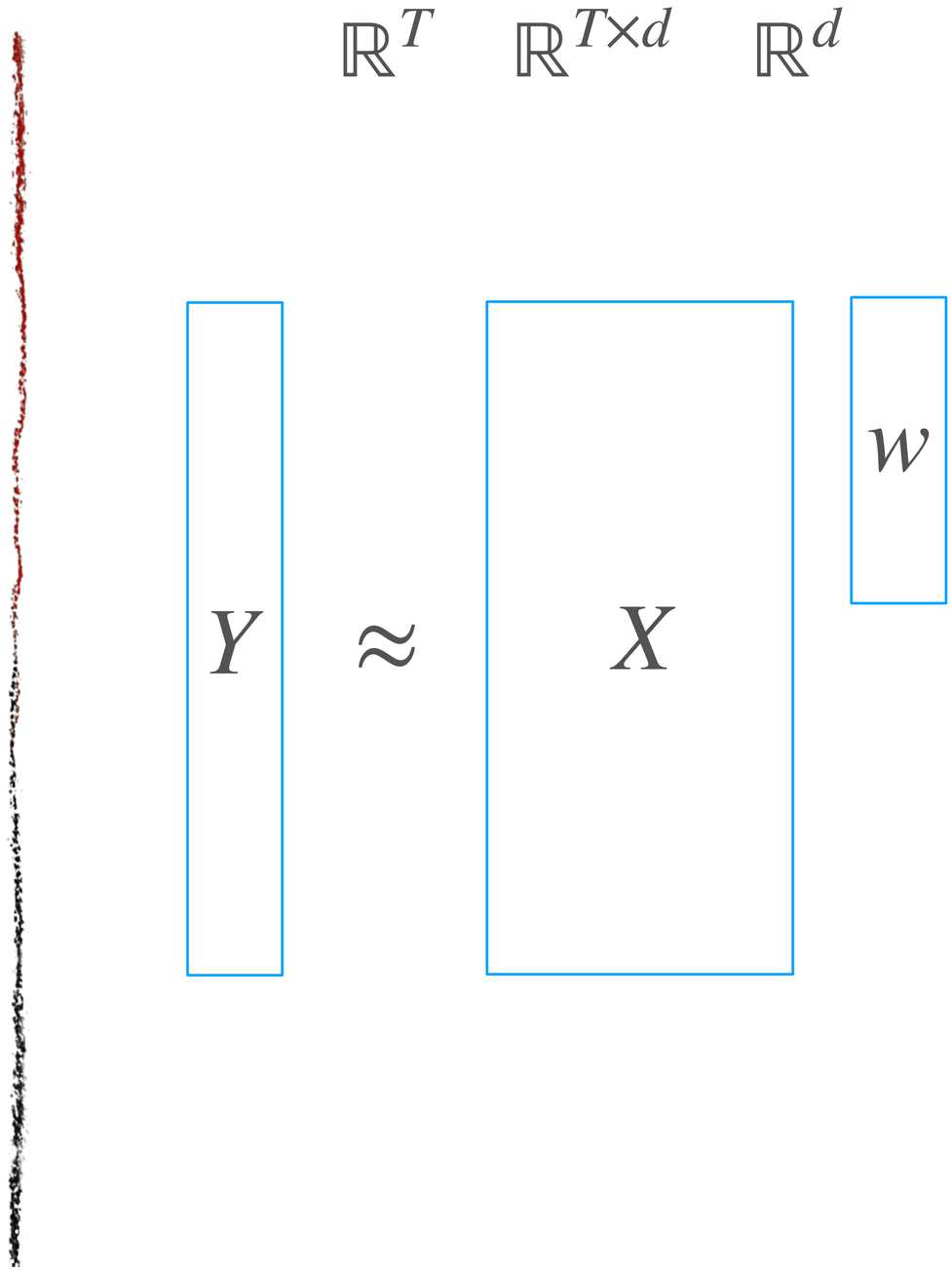


features X labels y

Family History	Resting Blood Pressure	Cholesterol	Cost (k$)
Yes	Low	Normal	0.1
No	Medium	Normal	0.2
No	Low	Abnormal	10.3
Yes	Medium	Normal	8.7
Yes	High	Abnormal	13.1
No	Low	Normal	0.1
No	Medium	Normal	0.4
Yes	Medium	Abnormal	9.6

- data matrix X, label vector Y; rows are $X^{(t)}$, $Y^{(t)}$

Schematic



features X labels y

	Family History	Resting Blood Pressure	Cholesterol	Cost (k$)
1	Yes	Low	Normal	0.1
1	No	Medium	Normal	0.2
1	No	Low	Abnormal	10.3
1	Yes	Medium	Normal	8.7
1	Yes	High	Abnormal	13.1
1	No	Low	Normal	0.1
1	No	Medium	Normal	0.4
1	Yes	Medium	Abnormal	9.6

- data matrix X, label vector Y; rows are $X^{(t)}$, $Y^{(t)}$

Schematic

$$\mathbb{R}^T \quad \mathbb{R}^{T\times d} \quad \mathbb{R}^d$$

$$Y \approx X \; w$$

$$L(w) = \|Y - Xw\|_F^2 \quad [+ R(w)]$$

↳ or 2

features X labels y

	Family History	Resting Blood Pressure	Cholesterol	Cost (k$)
1	Yes	Low	Normal	0.1
1	No	Medium	Normal	0.2
1	No	Low	Abnormal	10.3
1	Yes	Medium	Normal	8.7
1	Yes	High	Abnormal	13.1
1	No	Low	Normal	0.1
1	No	Medium	Normal	0.4
1	Yes	Medium	Abnormal	9.6

- data matrix X, label vector Y; rows are $X^{(t)}$, $Y^{(t)}$

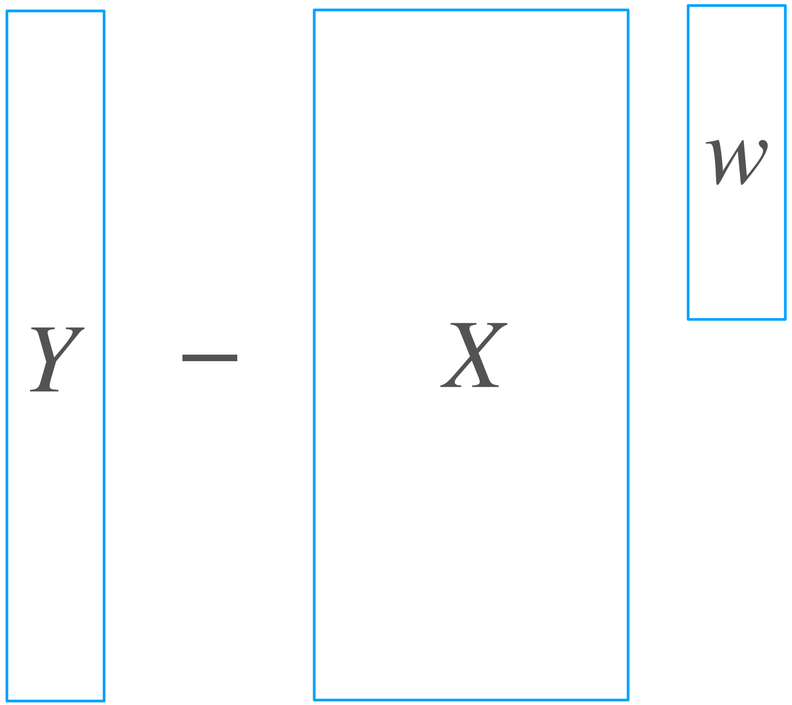
MOO!

- Previous slides illustrate *model, objective, optimizer* for linear regression
- Often three good questions to ask when learning a new ML method — maybe not complete, but a good place to start

MOO examples

- Linear regression
  - model = linear functions of features
  - objective = sum of squared errors (+ regularizer)
  - optimizer = exact or SGD
- ID3
  - model = decision trees
  - objective = mutual information
  - optimizer = greedy top-down induction

Gradients



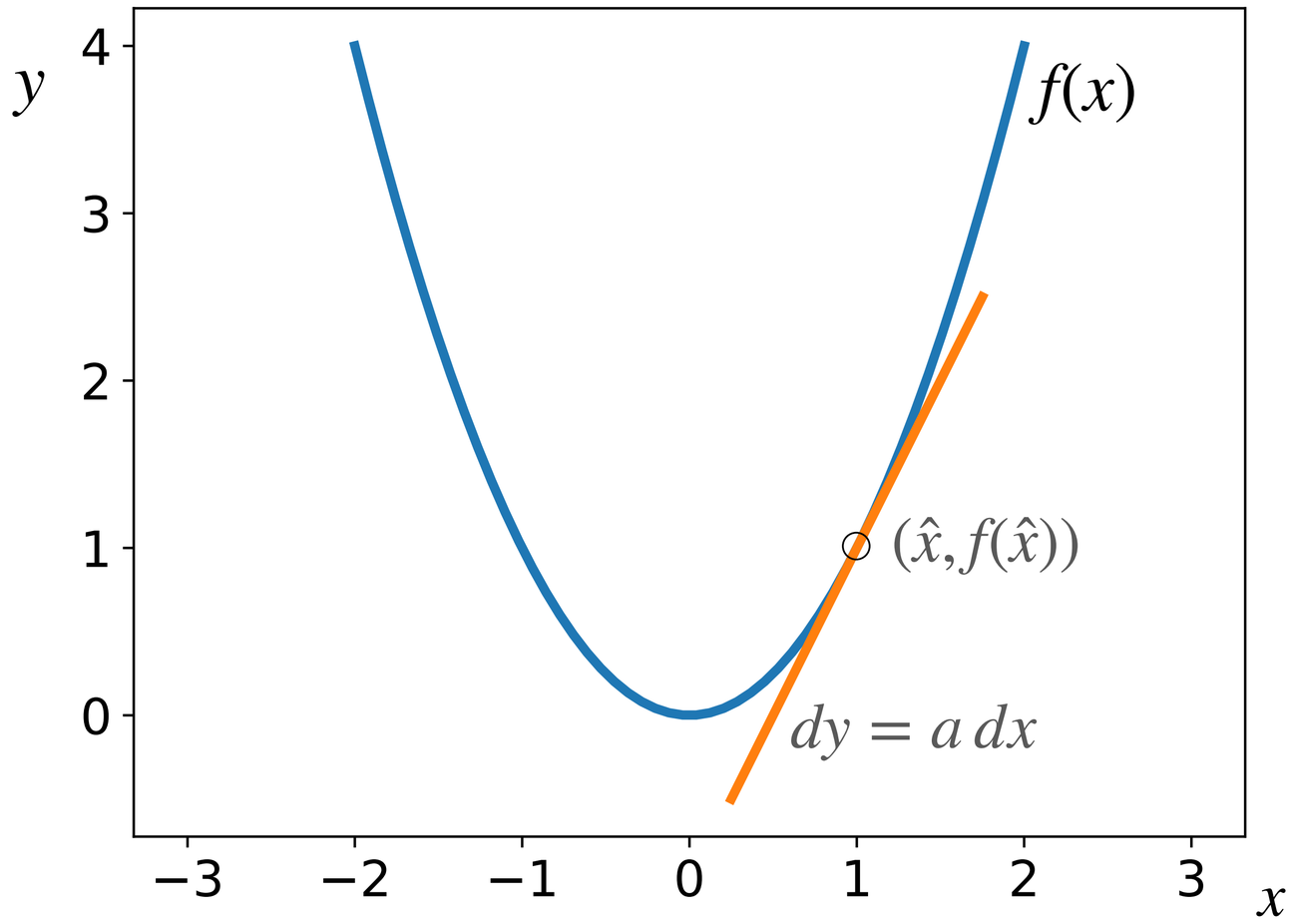
- For any of the above, need gradients
  - $L(w) = \frac{\lambda}{2}\|w\|^2 + \frac{1}{2}\sum_{t=1}^{T}(Y^{(t)} - w \cdot X^{(t)})^2$

    $= \frac{\lambda}{2}\|w\|^2 + \frac{1}{2}\|Y - Xw\|^2$
  - $\nabla L(w) =$?

Derivatives everywhere

- The way we derive many ML methods:
  - start from objective to minimize (here, sum of squared errors plus optional regularizer)
  - take a few derivatives ($0 = d\,\text{objective} / dw$)
  - find optimum (e.g., by setting to 0 and solving, or SGD)
  - done!
- Will continue to be a common pattern
- Review of intuition and tools for high-d derivatives, so we can do it fast

Derivatives notation



$m = n = 1$

$dy = y - f(\hat{x})$
$dx = x - \hat{x}$

dx, dy are called ***differentials***

- Function $y = f(x)$
  - $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$, $f \in \mathbb{R}^n \to \mathbb{R}^m$
- Derivative is a *linear fn* that *locally* approximates f near $\hat{x}$
  - $dy = a\,dx$, where $a \in \mathbb{R}$ is the derivative

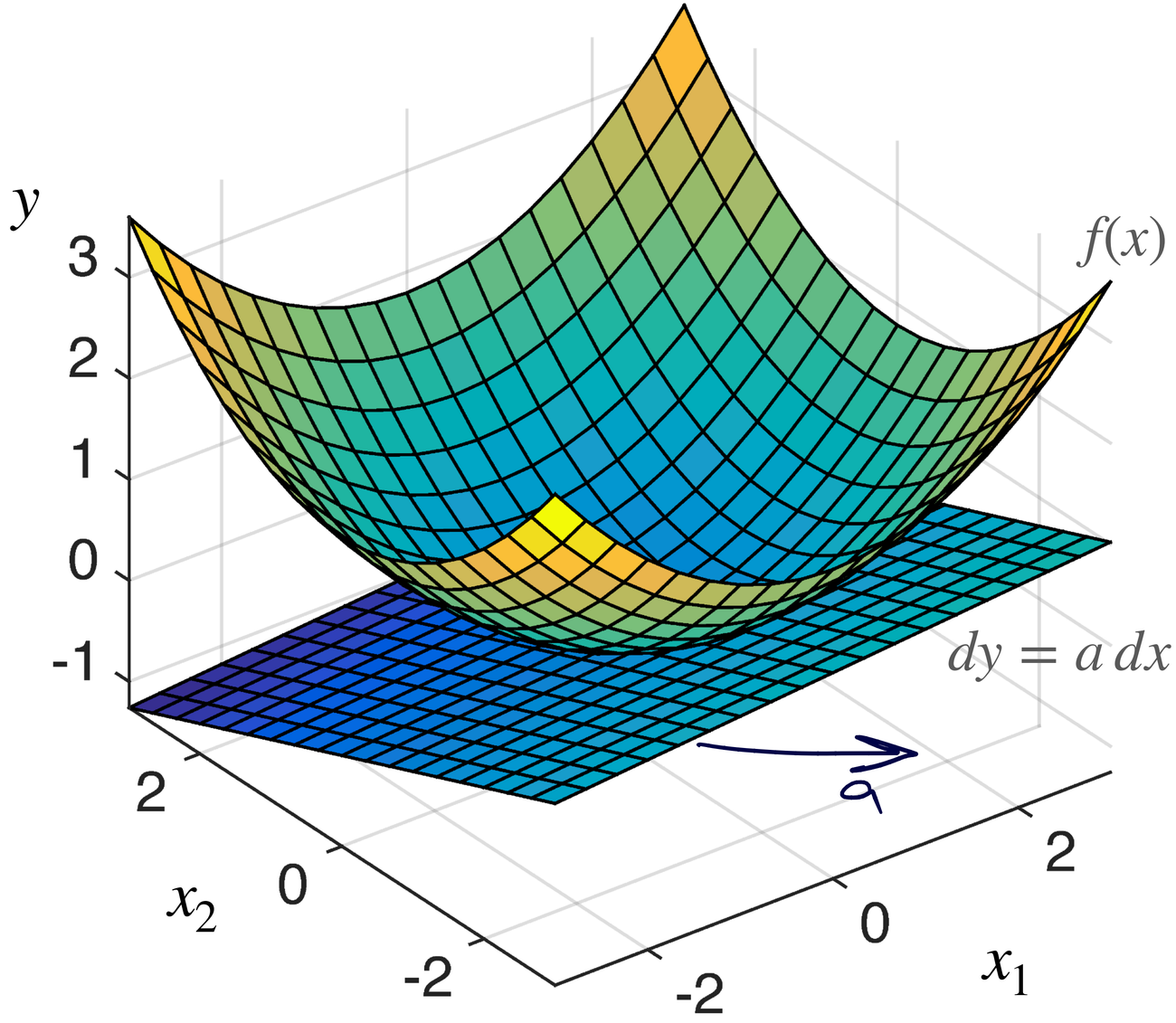
Compare older vs. newer notation

$dy = a\,dx$

$\frac{dy}{dx} = a$

- Older version:
  - $\frac{dy}{dx}$ is atomic: derivative of y wrt x
  - dx and dy are meaningless on their own
  - focused on infinitesimals: change in x or $y \to 0$
- Newer version:
  - dx and dy have separate meaning: terms in a linear function (the Taylor expansion)
  - dx and dy can be any size: linear function is well defined even for large changes in x or y (though accuracy of Taylor approximation can be bad if dx, dy are big)
- Heuristic to convert: “divide through” or “multiply through” by dx (only makes sense in scalar case, but often succeeds in converting notation anyway)

Higher dimensions

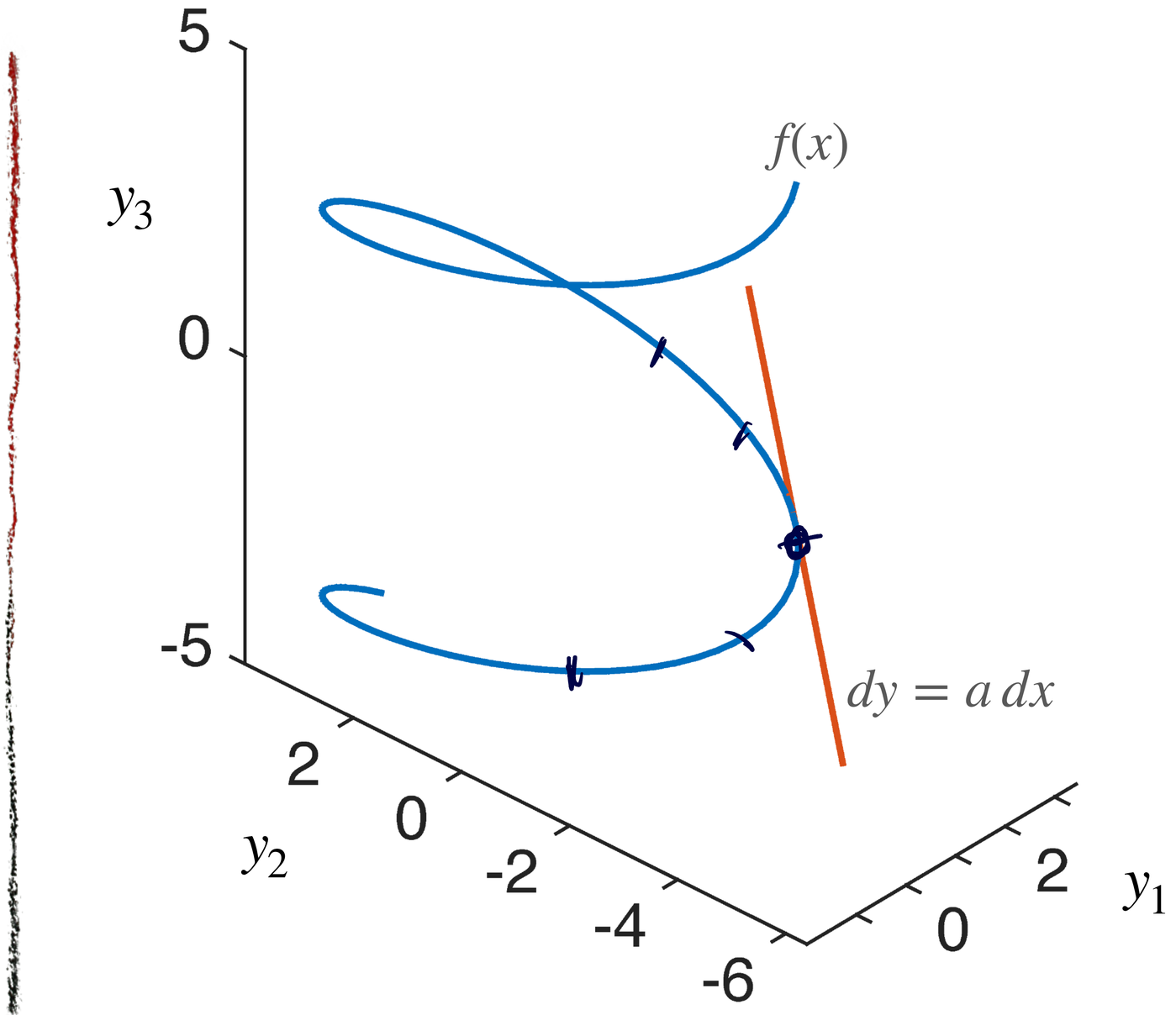


$\mathbb{R}^n \to \mathbb{R}^m$

$m = 1, n = 2$

Function $y = f(x)$ $\quad dy = a\,dx \quad a \in \mathbb{R}^{1\times 2}$

Higher dimensions



$\mathbb{R}^n \to \mathbb{R}^m$

$m = 3, n = 1$

Function $y = f(x)$ $\quad dy = a\,dx \quad a \in \mathbb{R}^{3\times 1}$

Notation for linear functions

output shape (derivative of) \ input shape (with respect to)	Scalar	Vector	Matrix
Scalar	$ds = a\,dt$	$ds = u^T dv$	$ds = \mathrm{tr}(M^T dN)$
Vector	$du = v\,ds$	$du = M\,dv$	$\times$
Matrix	$dM = N\,ds$	$\times$	$\times$

- a, s, t: scalars u, v: column vectors M, N: matrices
- note: $ds = a\,dt$ is a synonym for $a = \frac{ds}{dt}$, etc.
- $\times$ means no common notation; see torch.einsum

Notation for linear functions

output shape (derivative of) \ input shape (with respect to)	Scalar	Vector	Matrix
Scalar	$ds = a\,dt$	gradient	$ds = \mathrm{tr}(M^T dN)$
Vector	velocity	Jacobian	$\times$
Matrix	$dM = N\,ds$	$\times$	$\times$

- a, s, t: scalars u, v: column vectors M, N: matrices
- note: $ds = a\,dt$ is a synonym for $a = \frac{ds}{dt}$, etc.
- $\times$ means no common notation; see torch.einsum

Inner product

- With scalar output s, for any shape of input X, can write $ds = \langle A, dX \rangle$ where A, X, dX all have same shape
  - $\langle \cdot , \cdot \rangle$ is ***inner product***, in our case $\sum_{\text{index lists } I} A_I \, dX_I$
  - e.g., if A, dX are $m \times n$ matrices, then

    $$\langle A, dX \rangle = \sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij} \, dX_{ij}$$
- Expressions on previous slide are equivalent to above
  - most complex is $\text{tr}(M^\top dN)$
  - tr is sum of diagonal elements of a square matrix

    $$\left[M^\top dN\right]_{jk} =$$
  - sum of diagonal elements means set $j = k$, sum over j:

Shape conventions

- Derivative = coefficient of ds, dt, dv, dN on RHS
- Ambiguity: can write $\langle M, dN \rangle$ or $\mathrm{tr}(M^\top dN)$
  - is the derivative M or $M^\top$?
- Convention: x, y, and derivative $\frac{dy}{dx}$ are ***tensors*** (multidimensional arrays of numbers)
- In derivative $\frac{dy}{dx}$:
  - the dimensions of y come first, then x
  - in same order as dimensions of y and x
- For example: if y is 3×2 and x is $4 \times 6 \times 5$ then $\frac{dy}{dx}$ is $3 \times 2 \times 4 \times 6 \times 5$

Special cases

- If y is a scalar, then $\frac{dy}{dx}$ is the same shape as x
  - derivative of a scalar wrt a 5×3 matrix is a 5×3 matrix
- If x and y are vectors, $\frac{dy}{dx}$ (the Jacobian) is the same shape as a linear function from x to y
  - derivative of a 3-vector wrt a 6-vector is a 3×6 matrix

Useful identities

compatible dimensions:

- *2nd dimension of A = 1st dimension of B*
- *2nd dimension of B = 1st dimension of C*
- *2nd dimension of C = 1st dimension of A*

- scalar = scalar$^\top$ = tr(scalar)
- Trace rotation
  - tr(ABC) = tr(CAB) when A, B, C have compatible dimensions

- Transpose-Hadamard
  - $A \circ B =$ Hadamard (componentwise) product,

    $$[A \circ B]_{ij} = A_{ij} B_{ij}$$
  - $A^\top (B \circ C) = (A \circ B)^\top C$

Tools for derivatives: linearity, chain, product

- Linearity:
  - $d(ax + b) = a\,dx + \cancel{db}$, $d(AX + B) = A\,dX + \cancel{dB}$, …
- Chain rule for $f(g(x))$
  - old notation: $\frac{df}{dx} = \frac{df}{dg}\frac{dg}{dx}$ ("cancel the dg")
  - new: if $df = A\,dg$, $dg = B\,dx$ then $df = AB\,dx$
- Product rule for $f(x)\,g(x)$
  - $d(fg) = df\,g + f\,dg$
  - works for any kind of product (matrix product, vector cross product, elementwise (Hadamard) product, vector convolution, Khatri-Rao product …)
  - unlike scalar case, order can matter (products are not necessarily commutative)

Partial vs. total derivatives

- In a function of several variables, partial derivative imagines changing one of them, holding others fixed
  - e.g., $\frac{\partial}{\partial x}(x+y)^2 = 2(x+y)$ holding y fixed
- New notation: any variable that appears with d in RHS is changing, others are fixed
  - e.g., $d(x+y+z)^2 =$ holding z fixed

$$2(x+y+z)\underbrace{d(x+y+z)}_{(dx+dy)}$$

Example: chain rule with multiple variables

$du = \sin x \; dx$

$dv = 2x \, dx$

- Ex: $y = (u+v)^3, \;\; u = \cos x, \;\; v = x^2$

$$dy = 3(u+v)^2 (du + dv)$$
$$= 3(u+v)^2 (\sin x + 2x) \, dx$$

- Sometimes called ***law of total derivatives***:

$$\begin{aligned} df(u(x), v(x), w(x)) = {} & f^{(1)}(u, v, w) \, u'(x) \, dx \\ & + f^{(2)}(u, v, w) \, v'(x) \, dx \\ & + f^{(3)}(u, v, w) \, w'(x) \, dx \end{aligned}$$

Gradient practice: Linear regression in one slide

$$L = \frac{1}{2}\|Y - Xw\|^2 = (Y - Xw)^\top (Y - Xw)$$

$$L = \frac{1}{2}(Y^\top Y - 2Y^\top Xw + w^\top X^\top Xw)$$

$$dL = \frac{1}{2}(0 - 2Y^\top X\,dw + dw^\top X^\top X w + w^\top X^\top X\,dw)$$

$$dL = -Y^\top X\,dw + w^\top X^\top X\,dw$$

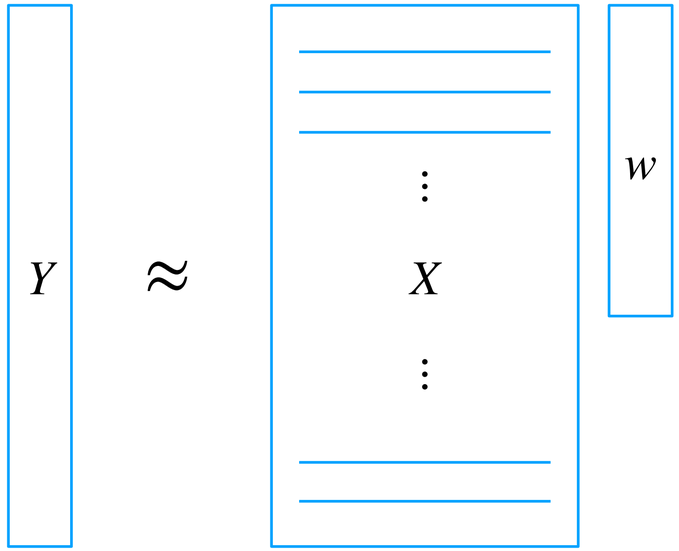
$$= (w^\top X^\top X - Y^\top X)\,dw$$

$$0 = X^\top X w - X^\top Y \qquad \boxed{X^\top X w = X^\top Y}$$

normal equations

Exact solution for ridge regression

$X \in \mathbb{R}^{T \times d}$
$Y \in \mathbb{R}^{T}$
$w \in \mathbb{R}^{d}$



- Predictions are $\hat{Y} = Xw$
- Objective is $\frac{\lambda}{2}\|w\|^2 + \frac{1}{2}\|Y - Xw\|^2$
- Gradient is $\lambda w + X^\top(Xw - Y)$
- Exact solution is $(\lambda I + X^\top X)w = X^\top Y$
  - implemented in `torch.linalg.lstsq`

or np.

Exact solution for ridge regression

$X \in \mathbb{R}^{T \times d}$

$Y \in \mathbb{R}^{T}$

$w \in \mathbb{R}^{d}$

- Pred
- Obje
- Grad
- Exac
  - implemented in `torch.linalg.lstsq`

Why use torch.linalg.lstsq? Handles (many) special cases: over- or under-determined, rank-deficient, poor conditioning

It would be several days of lectures to describe algorithms and best practices for all of these

Please remember: don't write inv() unless you are sure none of the special cases will bite you

https://forms.gle/hoenoXZA57MZ4GS3A

Poll

- In which of the following situations is selection bias the *largest*:
  - A: We roll 3 six-sided dice and take the largest
  - B: We roll 10 six-sided dice and take the largest
  - C: We roll 10 six-side dice and take the largest; the dice are weighted so that the mean is still $3\frac{1}{2}$, but extreme outcomes 1 & 6 have lower probability
  - D: We roll 15 six-sided dice; 5 of them are normal, but 10 are weighted so that they always come up 1 or 2
  - E: All the above are about the same

- Bias = $\mathbb{E}$(observed value of selected die) – $\mathbb{E}$(that same die if we roll it again)